FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

3

CASE-BASED REASONING (CBR) SYSTEM DEVELOPMENT

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I. BACKGROUND INFORMATION

Case-Based Reasoning (CBR) is an intelligent problem-solving approach used in **Knowledge Representation and Reasoning (KRR)**. Instead of relying solely on predefined rules, CBR systems solve new problems by referring to past experiences, also known as cases. These cases contain previous problems and their successful solutions. When a new situation arises, the system identifies the most similar past case, adapts the old solution to fit the new context, and applies it. Afterward, the new case is saved, allowing the system to learn and continuously improve.

CBR is widely used in various fields, such as disaster response, legal assistance, and intelligent tutoring, because it mimics how humans naturally make decisions. For example, emergency responders rely on past typhoon data to plan evacuations; lawyers examine previous legal cases to build arguments; and tutoring systems look at prior student mistakes to provide customized guidance. In all these examples, past knowledge becomes the foundation for solving future problems more effectively.

This activity allows students to experience how CBR works by developing a simple, functional CBR program based on real-world scenarios. Through pair programming, students will learn how to represent cases, implement similarity assessment, adapt solutions, and store new learning experiences—reflecting the complete CBR cycle. This hands-on approach strengthens understanding of how intelligent systems use experiential knowledge to support reasoning, decision-making, and learning.

II. TECHNICAL ASSESSMENT

CASE-BASED REASONING (CBR) SYSTEM DEVELOPMENT

Objective

At the end of this laboratory activity, students are expected to:

1. **Understand and apply the Case-Based Reasoning (CBR) cycle**, including retrieval, reuse, revision, and retention.
2. **Represent cases effectively** using appropriate data structures such as dictionaries, arrays, JSON objects, or classes.
3. **Implement a similarity assessment method** to identify relevant past cases based on defined attributes.
4. **Adapt retrieved solutions** to meet the needs of a new problem scenario based on the chosen case study.
5. **Demonstrate learning in CBR** by saving the newly solved case into the system's case base.

6. **Develop collaboration skills** through pair programming and joint problem analysis.
7. **Apply knowledge representation and reasoning concepts** to real-world problems in disaster response, legal reasoning, or intelligent tutoring.

Materials Needed

1. Hardware
   - A laptop or desktop computer
   - Stable internet connection (for research, documentation, submission)

2. Software

A programming environment:
   - Python (recommended: IDLE, VS Code, Jupyter Notebook, PyCharm)
   - Any text editor (Notepad++, Sublime, VS Code) if needed

3. File Formats & Tools

Word processor for documentation:
   - MS Word, Google Docs, or LibreOffice
   - Screenshot tool (e.g., Snipping Tool, Lightshot, Print Screen)

4. Reference Materials

Lecture notes on:
   - Knowledge Representation & Reasoning
   - Case-Based Reasoning cycle
   - Similarity measures

Provided case studies (Disaster DSS, Legal DSS, Intelligent Tutoring System)

5. Pair Collaboration Tools (optional but useful)
   - Google Drive for shared documentation
   - GitHub/GitLab for code collaboration

General Instructions

1. This **Technical Assessment (TA3)** must be completed by Pair.
2. Please ensure that you are grouped as pairs in the People tab.
3. Select only one case study from the list below.
4. Include clear comments in your code to describe its logic.
5. Each program should display the correct input, process, and output.
6. Ensure that your script displays clear and readable outputs.
7. Compile all Python codes and corresponding outputs into one PDF file.
8. Save it as **TA3_lastname1_lastname2.pdf**
   (Example: TA2_Patilano_Loverez.pdf)
9. Submit the **python file** and **PDF** file.

📌 Step-by-Step Instructions

1. Form a pair

Choose a partner. Each pair will work collaboratively on research, coding, debugging, and presenting your results.

2. Choose ONE (1) case study

Select only one case study from the list below. Your program must be based on the scenario and logic of the chosen system:

Case Study Options:

1. *Disaster Response Decision Support System (DSS)*
2. *AI-Based Legal Decision Support System*
3. *Intelligent Tutoring System (ITS)*

3. Define your case base

Create at least **3–5 past cases** stored as dictionaries, JSON objects, arrays, or any structure.

Example structure:

```
{
  "problem": {...},
  "solution": {...}
}
```

4. Implement similarity assessment

Your program must include a way to measure similarity between the new case and past cases.

Examples:

- Attribute matching
- Weighted scoring
- Simple rule-based comparison

Output should show: (sample)

✔ Similarity score or ranking

✔ The most relevant past case retrieved

5. Adapt the solution

Modify the retrieved solution so it fits the new problem scenario.

This part should reflect the logic of your chosen case study:

Examples:

- **DSS:** Adjust evacuation plans based on population size
- **Legal DSS:** Modify arguments due to new digital evidence
- **ITS:** Give personalized feedback based on student errors

6. Demonstrate learning (retain stage)

After generating a final solution, store the new case into your case base.

Your program must show:
✔ "New case saved" or similar confirmation
✔ Updated case list

7. Display clear program outputs

Your output must include:

- New problem description
- Retrieved similar case
- Adapted final solution
- Updated case base

8. Submission Requirements

Submit the following:

A. Program File

- Python

B. Short Documentation (1–2 pages)

Include:

- Title of chosen case study
- Description of your case base
- Explanation of similarity method
- Adaptation logic
- Screenshots of program execution

C. Pair Reflection (5–7 sentences)

Discuss:

- What each partner contributed
- What challenges you encountered
- What you learned about CBR

Case Studies:

Case Study 1: Disaster Response Decision Support System
A CBR-based DSS helps local government units respond to natural disasters. • Past cases include typhoons, floods, and earthquakes. • New event: *Typhoon with expected flooding in urban areas.* • System retrieves similar events: *Typhoon Ursula (2019)* and *Typhoon Odette (2021)*. • It adapts recommended responses such as evacuation strategy, resource allocation, and shelter capacity. • After execution, the system stores new outcomes for improved future reasoning.

Learning Outcome:
CBR helps governments respond faster by using *experiential knowledge* rather than rebuilding decisions from scratch.

Case Study 2: AI-Based Legal Decision Support System

A legal CBR tool assists lawyers by retrieving previous court cases similar to the current legal issue.
Scenario:
Client is charged with "theft involving electronic transfer." System retrieves past cases involving:

- online theft
- digital fraud
- unauthorized bank transfers

Adaptation:
The reasoning engine adjusts argument patterns to match the new technical context.
Learning:
After final verdict, the case is stored—enhancing the system's knowledge for future cybercrime-related offenses.

Case Study 3: Intelligent Tutoring System

A CBR system identifies the learning struggles of students.
Student submits a math solution with repeated errors.
System retrieves past students with similar error patterns (e.g., misusing formulas).
It adapts feedback and personalized lessons based on successful interventions.
Learning:
System improves by retaining new student profiles and outcomes.

Your Code - Laboratory Tasks ⇩

Source code:

```
# ========================================
# CASE-BASED REASONING (CBR)
# Intelligent Tutoring System
# ========================================

# 1. CASE BASE
# These are previous student cases.
case_base = [
    {
        "topic": "Algebra",
        "error": "wrong formula",
        "solution": "Review the correct algebra formula."
    },
    {
```

```
        "topic": "Algebra",
        "error": "calculation error",
        "solution": "Practice basic calculations."
    },
    {
        "topic": "Geometry",
        "error": "wrong formula",
        "solution": "Review the correct geometry formula."
    },
    {
        "topic": "Geometry",
        "error": "calculation error",
        "solution": "Practice geometry calculations."
    }
]


# 2. NEW PROBLEM
print("=== INTELLIGENT TUTORING SYSTEM ===")

topic = input("Enter topic (Algebra/Geometry): ")
error = input("Enter error (wrong formula/calculation error): ")

new_case = {
    "topic": topic,
    "error": error
}


# 3. RETRIEVE
# Find the most similar previous case.
best_case = None
highest_score = -1

for case in case_base:

    score = 0

    if case["topic"].lower() == topic.lower():
        score += 1

    if case["error"].lower() == error.lower():
        score += 1
```

```
    print("Similarity Score:", score)

    if score > highest_score:
        highest_score = score
        best_case = case


# Display retrieved case
print("\n=== RETRIEVED CASE ===")
print("Topic:", best_case["topic"])
print("Error:", best_case["error"])
print("Similarity Score:", highest_score)


# 4. REUSE AND ADAPT
print("\n=== ADAPTED SOLUTION ===")

if error.lower() == "wrong formula":
     solution = "Review the correct formula and practice similar
problems."
else:
      solution = "Practice solving problems carefully and check
your calculations."

print(solution)


# 5. RETAIN
# Save the new case to the case base.
new_case["solution"] = solution
case_base.append(new_case)

print("\n=== RETENTION ===")
print("New case saved!")


# 6. UPDATED CASE BASE
print("\n=== UPDATED CASE BASE ===")

for case in case_base:
    print(case)
```

Output:

```
=== INTELLIGENT TUTORING SYSTEM ===
Enter topic (Algebra/Geometry): algebra
Enter error (wrong formula/calculation error): wrong formula
Similarity Score: 2
Similarity Score: 1
Similarity Score: 1
Similarity Score: 0

=== RETRIEVED CASE ===
Topic: Algebra
Error: wrong formula
Similarity Score: 2

=== ADAPTED SOLUTION ===
Review the correct formula and practice similar problems.

=== RETENTION ===
New case saved!

=== UPDATED CASE BASE ===
{'topic': 'Algebra', 'error': 'wrong formula', 'solution': 'Review the correct algebra formula.'}
{'topic': 'Algebra', 'error': 'calculation error', 'solution': 'Practice basic calculations.'}
{'topic': 'Geometry', 'error': 'wrong formula', 'solution': 'Review the correct geometry formula.'}
{'topic': 'Geometry', 'error': 'calculation error', 'solution': 'Practice geometry calculations.'}
{'topic': 'algebra', 'error': 'wrong formula', 'solution': 'Review the correct formula and practice similar problems.'}
PS C:\Users\Dom>
```

```
=== INTELLIGENT TUTORING SYSTEM ===
Enter topic (Algebra/Geometry): geometry
Enter error (wrong formula/calculation error): calculation error
Similarity Score: 0
Similarity Score: 1
Similarity Score: 1
Similarity Score: 2

=== RETRIEVED CASE ===
Topic: Geometry
Error: calculation error
Similarity Score: 2

=== ADAPTED SOLUTION ===
Practice solving problems carefully and check your calculations.

=== RETENTION ===
New case saved!

=== UPDATED CASE BASE ===
{'topic': 'Algebra', 'error': 'wrong formula', 'solution': 'Review the correct algebra formula.'}
{'topic': 'Algebra', 'error': 'calculation error', 'solution': 'Practice basic calculations.'}
{'topic': 'Geometry', 'error': 'wrong formula', 'solution': 'Review the correct geometry formula.'}
{'topic': 'Geometry', 'error': 'calculation error', 'solution': 'Practice geometry calculations.'}
{'topic': 'geometry', 'error': 'calculation error', 'solution': 'Practice solving problems carefully and check your calculations.'}
PS C:\Users\Dom>
```

```
=== INTELLIGENT TUTORING SYSTEM ===
Enter topic (Algebra/Geometry): algebra
Enter error (wrong formula/calculation error): calculation error
Similarity Score: 1
Similarity Score: 2
Similarity Score: 0
Similarity Score: 1

=== RETRIEVED CASE ===
Topic: Algebra
Error: calculation error
Similarity Score: 2

=== ADAPTED SOLUTION ===
Practice solving problems carefully and check your calculations.

=== RETENTION ===
New case saved!

=== UPDATED CASE BASE ===
{'topic': 'Algebra', 'error': 'wrong formula', 'solution': 'Review the correct algebra formula.'}
{'topic': 'Algebra', 'error': 'calculation error', 'solution': 'Practice basic calculations.'}
{'topic': 'Geometry', 'error': 'wrong formula', 'solution': 'Review the correct geometry formula.'}
{'topic': 'Geometry', 'error': 'calculation error', 'solution': 'Practice geometry calculations.'}
{'topic': 'algebra', 'error': 'calculation error', 'solution': 'Practice solving problems carefully and check your calculations.'}
```

```
PS C:\Users\Dom> python -u "d:\Feu Alabangseu SY 26 - 27 4th year\first sem\CS Elective - Intelligent Systems\Lab act\Technic
=== INTELLIGENT TUTORING SYSTEM ===
Enter topic (Algebra/Geometry): geometry
Enter error (wrong formula/calculation error): wrong formula
Similarity Score: 1
Similarity Score: 0
Similarity Score: 2
Similarity Score: 1

=== RETRIEVED CASE ===
Topic: Geometry
Error: wrong formula
Similarity Score: 2

=== ADAPTED SOLUTION ===
Review the correct formula and practice similar problems.

=== RETENTION ===
New case saved!

=== UPDATED CASE BASE ===
{'topic': 'Algebra', 'error': 'wrong formula', 'solution': 'Review the correct algebra formula.'}
{'topic': 'Algebra', 'error': 'calculation error', 'solution': 'Practice basic calculations.'}
{'topic': 'Geometry', 'error': 'wrong formula', 'solution': 'Review the correct geometry formula.'}
{'topic': 'Geometry', 'error': 'calculation error', 'solution': 'Practice geometry calculations.'}
{'topic': 'geometry', 'error': 'wrong formula', 'solution': 'Review the correct formula and practice similar problems.'}
PS C:\Users\Dom>
```

B. Short Documentation (1–2 pages) Include:

- **Title of chosen case study**

- **Case-Based Reasoning (CBR) Intelligent Tutoring System**

- **Description of your case base**

- **Our case base contains examples of previous student mistakes in Algebra and Geometry. Each case includes the topic, the type of error made by the student, and a suggested solution. We used these previous cases as references so that the system can find a similar problem when a new student case is entered. This shows how CBR can use past experiences to help solve new problems.**

- **Explanation of similarity method**

- **The system checks how similar the new problem is to the cases already stored in the case base. It compares two things: the topic and the type of error. If the topic matches, the case gets 1 point, and if the error also matches, it gets another 1 point. The case with the highest score is considered the most similar case and is selected by the system.**

- **Adaptation logic**

- **After finding the most similar case, the system provides a solution based on the student's error. If the student made a wrong formula error, the system suggests reviewing the correct formula and practicing similar problems. If the student made a calculation error, the system suggests practicing calculations and checking the answers carefully. This allows the system to adjust the previous solution to fit the new problem.**

- **Screenshots of program execution**

C. Pair Reflection (5–7 sentences)

Discuss:

- What each partner contributed
- What challenges you encountered
- What you learned about CBR

Pair Reflection: My partner and I worked together in creating, testing, and documenting our Case-Based Reasoning system. We divided the tasks so that we could work more efficiently, with one of us focusing more on the case base and program logic while the other helped with testing, checking the results, and preparing the documentation. For the documentation, we worked together in explaining the case base, similarity method, adaptation logic, and the screenshots of the program execution. One challenge we encountered was making sure that the system could correctly compare the new case with the previous cases and choose the most similar one. We also had to check the conditions for different errors so that the system would give an appropriate solution. Through this activity, we learned that CBR uses previous experiences or cases to help solve new problems instead of starting from scratch. Overall, this activity helped us understand the retrieve, reuse, adapt, and retain steps of CBR and how they can be applied to create a simple Intelligent Tutoring System.